

SENIOR FULL STACK CHALLENGE · TURBO AI

Turbo Notes

A cozy little home for
your charming notes —
with AI voice.

 Django 5 · DRF · Postgres | Next.js 16 · TypeScript · TanStack Query

 Live · notes.cardenas.pe

✓ Backend 100% · 124 FE tests

✓ Playwright E2E

 httpOnly-cookie auth

OWASP Top 10

01 · THE BUILD

What it is

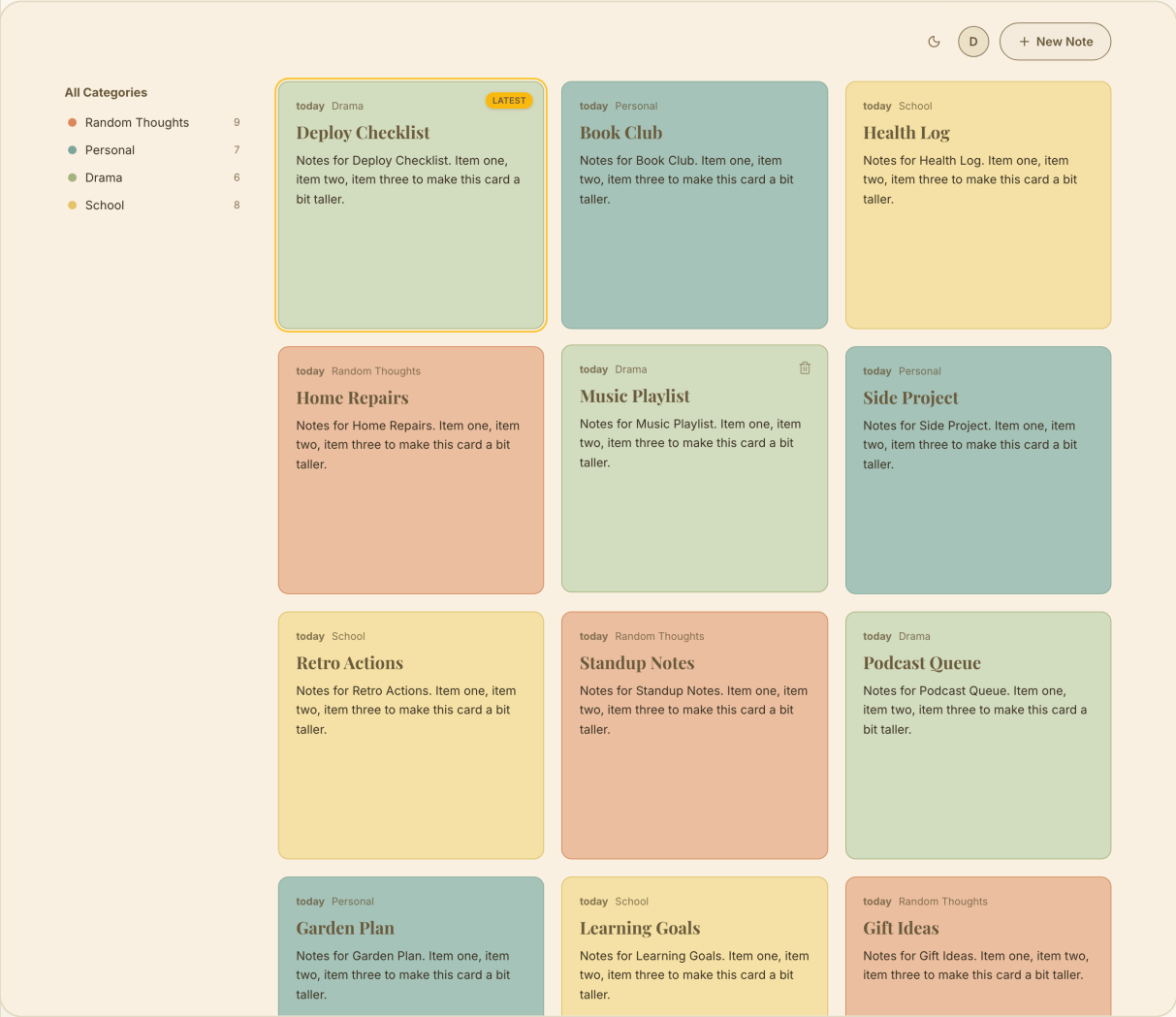
- **Per-user notes** — JWT auth, owner-scoped data (404 not 403, no existence leaks).
- **Autosave editor** — no save button; born on the first keystroke, PATCHed on an 800 ms debounce.
- **Color categories** — instant recolor; sidebar counts & filters.
- **AI, live & optional** — Whisper dictation · TTS read-aloud · suggest-title / summarize · hands-free "**close my note**".
- **Graceful degradation** — no key → free in-browser speech. AI is never a single point of failure.

Guiding principle

The right amount of engineering for the scope — clean layering, full coverage, documented tradeoffs, **no invented complexity**.

02 · SEE IT RUNNING

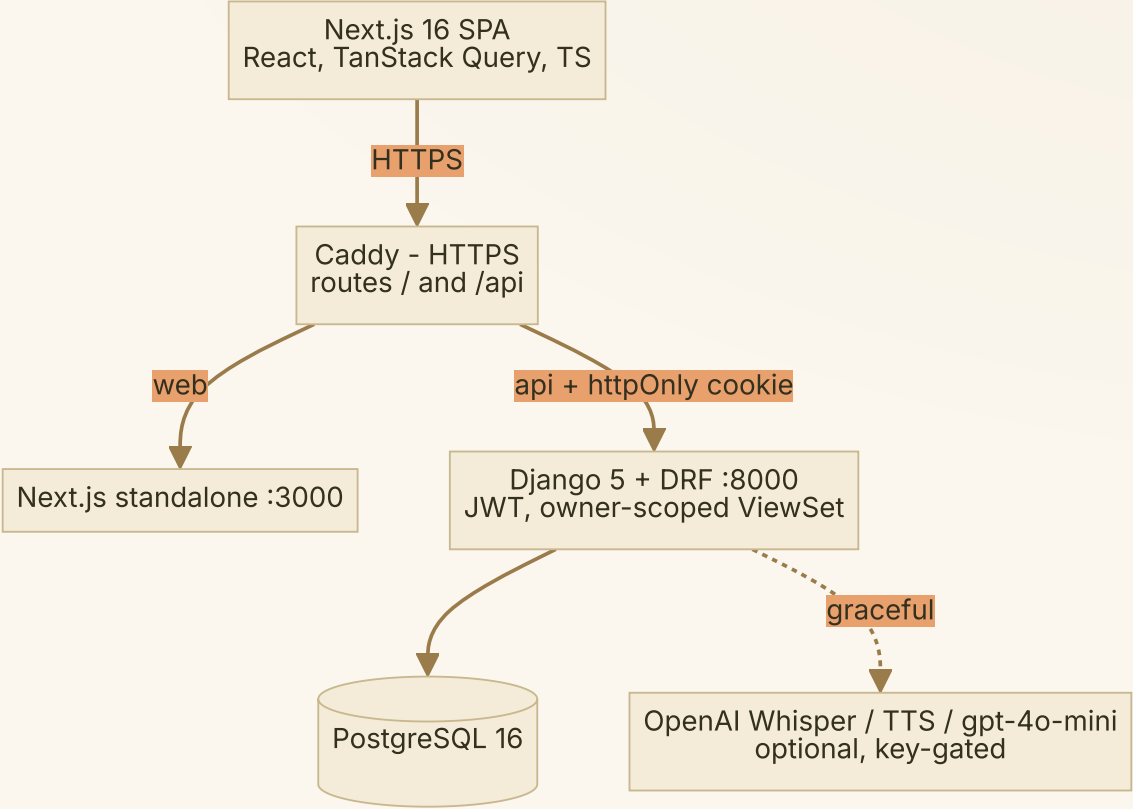
Live demo



-  **notes.cardenas.pe**
- Login `demo@turbo.ai` / `demo12345`
- Create a note → it **autosaves**; change category → recolors.
- Dictate by mic · read aloud · suggest title / summarize.
- Say "**close my note**" → AI names it, editor evaporates.
- Uniform card grid · newest note flagged `LATEST` · dark mode · infinite scroll.

03 · HOW IT FITS TOGETHER

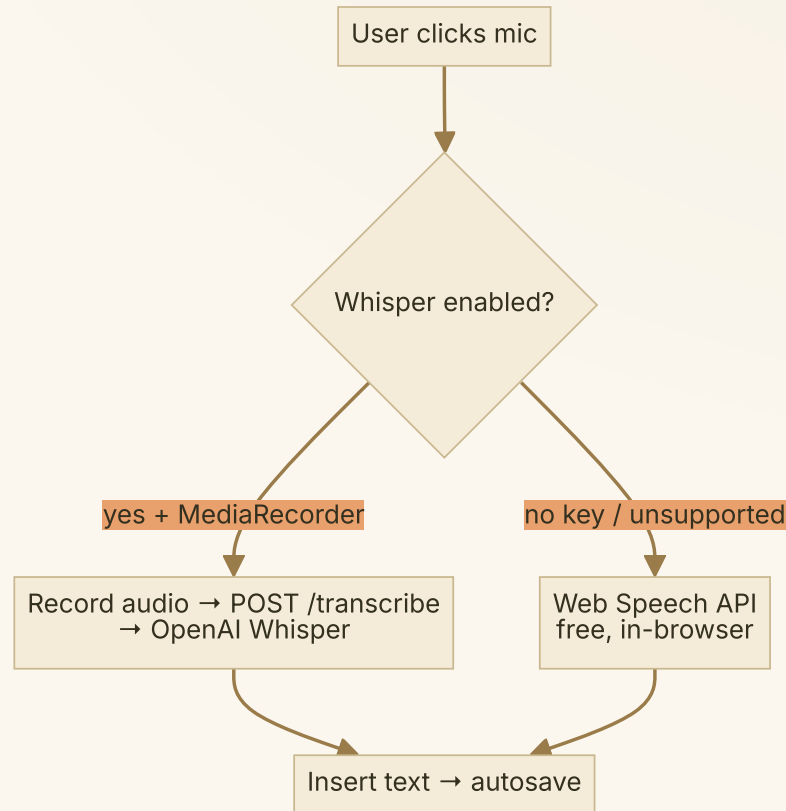
Architecture



Idiomatic DRF (ViewSet + serializers + models). Frontend layered `services → hooks → types` — mockable, unit-testable.

04 · A FLOW THAT SHOWS THE DESIGN

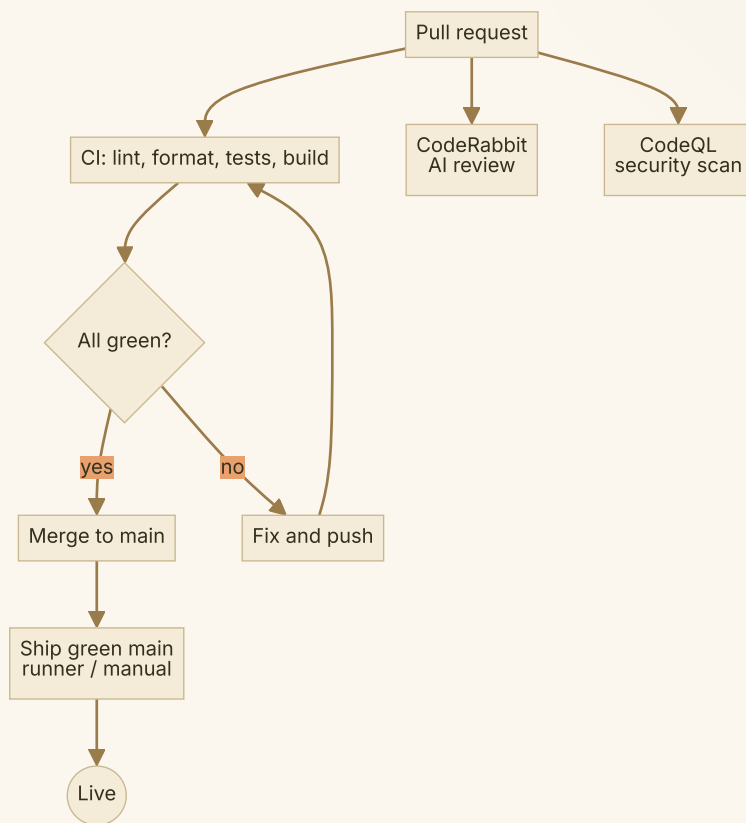
Voice → text, never a SPOF



Same pattern for TTS, assist, and the hands-free command — every AI path falls back to a free one. On a 401 the axios layer does a single transparent **cookie refresh**, then replays the request.

05 · ALWAYS GREEN BEFORE MERGE

CI/CD & quality



- **Gate** — flake8 · black/isort · pytest $\geq 85\%$ (now 100%) · Jest · next build.
- **CodeRabbit** — free AI PR review.
- **CodeQL** — TS + Python security scan.
- **Dependabot** — weekly deps + security PRs.

06 · SHIPPED, SAFELY

Hardened deploy & honest scope

"Saneado" deploy

- **Prod is never touched by a PR** — only reviewed, green `main` ships.
- Isolated Compose on `127.0.0.1:3300` · idempotent Caddy with **validate before reload**.
- Post-deploy `/api/health` check · secrets only on the box's `.env`.
- VPS firewalls inbound SSH → ships via **manual tar+ssh / self-hosted runner** (honest, not auto).

On the `k8s/` folder

Kubernetes manifests are **documentation of the scale-out path, not the deployment in use**. The app runs as one isolated Compose stack — the right size today. The seam to Deployment + Service + HPA is shown, **without overbuilding now**.

Testing: backend **pytest 100%** (CI floor 85%, all OpenAI calls mocked) · frontend **124 Jest tests** · **Playwright E2E** over the stack. ● Security: httpOnly-cookie auth, rate limiting, HSTS, audit log, OpenAI usage tracking, OWASP Top 10 write-up.

07 · ACTED ON THE REVIEW FEEDBACK

Post-challenge hardening

-  **httpOnly-cookie auth** — tokens left `localStorage`; now `HttpOnly · Secure · SameSite=Lax` cookies (no JS-readable token → no XSS exfiltration). Same-origin API ⇒ **Lax is the CSRF defence**, no extra token. Bearer still accepted for API clients.
-  **Refactored the editor** — the large `NoteEditor` split into focused, unit-tested hooks `useAiAssist · useReadAloud · useVoiceDictation` (1265 → ~840 lines, 97% cov).
-  **E2E with Playwright** — register → create → autosave-persists → delete, over the real stack, wired into CI. It caught a real auth-redirect bug.

Why it matters

Each item shipped as its own **PR → CI (incl. E2E) → merge → deploy**. The point isn't just the diff — it's **responding to feedback with verified, reviewable change**, and knowing **when SameSite alone is enough** vs. reaching for more.